

Polynomial Regression with Scikit-Learn

IEEE SSCS AUSC — AI Sub-Team · Task 6

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Dataset: UCI Auto MPG (+ synthetic quadratic check)

1. Objective

This report summarizes a full polynomial-regression workflow on the UCI Auto MPG dataset: predict miles per gallon (MPG) from vehicle attributes, compare a linear baseline against polynomial models (degrees 2–5), apply Ridge/Lasso regularization, and interpret the five core regression metrics plus residual diagnostics.

2. Methodology

- **Data:** 398 cars; 6 rows with missing horsepower (?) removed → 392 samples. Features: cylinders, displacement, horsepower, weight, acceleration, model year, origin.
- **Split:** 70% train / 15% validation / 15% test (random_state=42). Test used only for final scoring.
- **Pipeline:** PolynomialFeatures → StandardScaler → estimator to keep transforms leak-free.
- **Selection:** best unregularized degree chosen by validation RMSE; Ridge/Lasso alpha tuned with 5-fold GridSearchCV on train+validation.

3. Results

3.1 Degree selection (train vs validation)

Degree	# Features	Train RMSE	Val RMSE	Train R²	Val R²
1	7	3.37	3.27	0.822	0.809
2	35	2.52	2.60	0.901	0.879
3	119	1.75	11.42	0.952	-1.33
4	329	0.00	177.14	1.000	≪ 0
5	791	0.00	72.89	1.000	≪ 0

Best unregularized degree = 2. Degrees ≥ 3 explode the feature space and overfit hard (train RMSE collapses while validation RMSE rises). This is a textbook bias–variance story.

3.2 Regularization

- **Ridge (L2)**: best $\alpha \approx 0.1$, CV RMSE ≈ 2.84
- **Lasso (L1)**: best $\alpha \approx 0.00178$, CV RMSE ≈ 2.83

3.3 Test-set comparison

Model	MSE	MAE	RMSE	MAPE (%)	R ²
Lasso (deg 2)	7.47	2.07	2.73	9.23	0.849
Ridge (deg 2)	7.68	2.10	2.77	9.34	0.845
Poly deg 2	7.97	2.12	2.82	9.51	0.839
Linear (deg 1)	9.10	2.21	3.02	11.02	0.816
Poly deg 3	12.47	2.69	3.53	13.35	0.748
Poly deg 4–5	severe overfitting — test metrics collapse				

Best model: **Lasso, degree 2** — about **9.4% lower test RMSE** than the linear baseline (2.73 vs 3.02 MPG).

4. Ridge & Lasso — Why They Matter for Polynomial Models

Polynomial expansion creates many correlated terms (e.g. weight^2 , $\text{weight} \times \text{horsepower}$). Without a penalty, least squares can assign huge coefficients that fit noise.

- **Ridge (L2)** shrinks coefficients toward zero but keeps all features. It is stable when predictors are multicollinear — common after polynomial expansion.
- **Lasso (L1)** can drive some coefficients exactly to zero, performing automatic feature selection. That is useful when only a subset of interaction/power terms matter.

In this task, both improved on unregularized degree-2, and Lasso edged Ridge on the held-out test set. Regularization is not optional decoration for high-degree polynomials — it is what keeps the model usable.

5. Regression Metrics — When to Use Each

Metric	What it measures	Use when...	Limitation
MSE	Mean of squared residuals	Large errors are especially costly; convenient loss for optimization	Not in original y-units; dominated by outliers
MAE	Mean absolute residual	You want a robust, easy-to-explain average error (MPG here)	Treats all errors linearly — may under-penalize rare large misses
RMSE			

	$\sqrt{\text{MSE}}$ — error in y-units, still outlier-sensitive	Primary reporting metric for this problem (interpretable MPG error)	Still pulled by outliers more than MAE
MAPE	Mean absolute % error	Comparing relative accuracy across scales	Breaks / blows up when true $y \approx 0$; asymmetric
R²	Fraction of variance explained vs predicting the mean	Quick relative model-quality score (0–1 typically)	Always rises with more features on train; can be negative on bad models
Adj. R²	R ² penalized for number of predictors	Comparing models with different feature counts fairly	Undefined / misleading when $\text{\#features} \geq \text{\#samples} - 1$ (high poly degrees)

For Auto MPG, **RMSE + R²** were the main pair: RMSE gives the typical miss in MPG, R² gives relative quality. MAPE (~9% for the best model) is a helpful secondary view; MAE is the most stakeholder-friendly absolute error.

6. Residual Analysis

For the best Lasso model, residuals ($\hat{y} - y$) were plotted against fitted values. The cloud is roughly centered near zero (mean residual ≈ 0) with no strong funnel shape, suggesting heteroscedasticity is not severe. A mild spread at higher fitted MPG values is visible — the model is slightly less precise on high-efficiency cars — but residuals do not show a clear systematic curve, which supports the degree-2 choice.

7. Conclusions

1. **Nonlinearity is real:** MPG vs horsepower/weight curves are not straight lines; a degree-2 expansion meaningfully beats a linear baseline.
2. **More degree is not more accuracy:** degrees 4–5 memorized the training set (train RMSE $\rightarrow 0$) and failed on validation/test — classic overfitting.
3. **Regularization finishes the job:** Lasso at degree 2 with a small tuned α delivered the best test RMSE (~2.73 MPG, $R^2 \approx 0.85$), showing the data's complexity is moderate — curved, but not arbitrarily high-order.

A synthetic quadratic check in the notebook confirms the same lesson: when the true process is quadratic, degree 2 recovers it and degree 1 underfits — validation curves, not training scores, decide the right complexity.